Surabhi Chavan

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EDUCATION

Master of Science in Information Systems San Diego State University, USA | Aug 2024 - May 2026 | GPA: 3.5
Bachelor of Engineering Savitribai Phule Pune University, India | May 2018 - Apr 2022 | GPA: 3.8

EXPERIENCE

Software Development Intern **Jan 2026 - Present**
Donald Hans Los Angeles, CA

- Delivered a production-ready AI portfolio assistant using React and TypeScript on Cloudflare Pages/Workers, enabling real-time LLM-powered edge conversations with sub-100ms response latency.
- Boosted deployment reliability 30% via async error-handling pipelines and multi-env debug workflows on Cloudflare edge.
- Enhanced SEO of production web properties by optimizing structured metadata, page speed, and Core Web Vitals signals.

Tech stack: React, Cloudflare, REST APIs, Agentic AI, SEO Optimization

Software Research Assistant **Jan 2025 - Dec 2025**
AI4Business - San Diego State University San Diego, CA

- Reduced LLM hallucination by 14% by developing a novel approach leveraging Agentic AI systems and AutoGen frameworks.
- Improved handwriting recognition accuracy to 92% by designing a supervised ML model trained on diverse datasets.
- Achieved 90% image classification accuracy by fine-tuning Deep Learning models (AlexNet CNN) in MATLAB.
- Enhanced email security by 30% through an NLP phishing-detection model that analyzes email pattern anomalies.

Tech stack: Python, PySpark, MATLAB, TensorFlow, Scikit-learn, AutoGen, Agentic AI, NLP, Data Analysis, Model Evaluation

Software Engineer **Sep 2022 - Aug 2024**
JSat Automation Inc. India

- Built a scalable scheduling system with Angular, REST APIs, and AWS Lambda/API Gateway for high-availability planning.
- Optimized real-time system performance by reducing latency 35% and improving usability 60% via TypeScript/RxJS reactive pipelines, refining async execution paths, and using CloudWatch metrics.
- Reduced build times by 40% through Gulp automation for transpilation, CSS preprocessing, and image pipeline enhancements.
- Boosted PostgreSQL query performance by 45% through indexing, plan tuning, and refined connection-pool management.
- Improved scalability by deploying Dockerized microservices with centralized logging and environment-specific configurations.
- Designed e-workflow and scheduling UX prototypes in Figma, enabling cross-functional stakeholder alignment before development, reducing rework cycles by 25%.

Tech stack: JavaScript (ES6+), TypeScript, Angular, RxJS, Node.js, REST APIs, PostgreSQL, Three.js, WebGL, Docker, AWS Lambda/API Gateway, S3, CloudWatch, Gulp, Figma, Git, Agile Development, CI/CD, UI/UX Design

Software Engineer Intern **Mar 2022 - Aug 2022**
Qualys Security India

- Improved vulnerability detection accuracy by 40% by automating CVE-CPE mapping through CISA API integration and repository synchronization.
- Built automation using the Shodan API to identify affected hosts and IPs, strengthening external attack surface monitoring.
- Developed scripts to compare technology repositories and generate CSV reports for missing CPEs and outdated entries.

Tech stack: Java, Spring Boot, REST APIs, JSON, Python, Oracle DB, Shodan API, CVE/CPE Mapping, Vulnerability Management

PROJECTS

ROOM | Video Conferencing Platform | React, Next.js, TypeScript, Clerk, getstream, shadcn, Tailwind CSS

- Built a real-time video conferencing platform using Next.js 14 and TypeScript for high performance and scalability.
- Integrated Stream Video SDK for live audio/video streaming and Clerk for secure authentication and user management.
- Delivered core features like instant meetings, scheduling, recordings, and join-via-ID to enhance usability.
- Designed an accessible, modern UI using shadcn/ui and Radix UI, enabling seamless collaboration for remote teams.

AI for Finance | XGBoost, Python, AWS SageMaker, Pandas, NumPy, Matplotlib

- Achieved 92% fraud detection accuracy by building and tuning an XGBoost classifier on large-scale transactional datasets.
- Reduced false positives by 15% through systematic feature engineering, regularization tuning (max_depth, gamma, eta), and model evaluation via confusion matrix.
- Optimized training efficiency by leveraging AWS SageMaker for distributed CPU computation and scalable model deployment.
- Enhanced model interpretability and transparency using feature importance plots and Matplotlib-driven performance insights.